CAMARINES SUR POLYTECHNIC COLLEGES
COLLEGE of COMPUTER STUDIES

CSAC 224 - Machine Learning

Evaluating the Impact of Study Time on Student's Academic Performance: A Univariate Linear Regression Approach

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Abstract

Students Academic Performance is affected by different factors including the 'Study Time'. This study investigates the importance of having enough study time on students and how it affect their academic success. A Univariate Linear Regression is used in this study to evaluate the relationship between study time and student's performance. A dataset gathered from Kaggle repository contains 2 columns which are "Hours" and "Scores" that is used as the input feature for training and assessing the model's performance. This study includes flask application to provide a user friendly interface where a specific number of hours will be entered and the model will evaluate and predict the students performance. The model performed well with an accuracy of 94.55% implying that the number of study hours significantly affect a student's performance.

1 Introduction

Academic Achievement has been an important goal for every students in this generation, they strive for good grades and performs excellently in class in order to ensure a success on their educational journeys and for a so called 'Academic Validation'. Educators help students on having this by imparting knowledge to them and enhancing their skills. However, some student shows poorer performance than the others despite of having a similar learning method, There are numerous factors for this and one of those is the time spent by the students studying.

This study evaluates how the students study time affect their academic performance. a univariate linear regression will be used to analyze the importance of having enough study time and if having it does really affect their performance in school. This study is relevant today as most of the students are having a hard time managing their time for studies and other activities they have to attend. Conducting this study will give a better understanding to the students on how study time influence their academic success.

2 Related work

In a study conducted by Yazan Alsariera, et al., (2022) a student performance is analyzed using different variables and various machine learning algorithms which are KNN classifier, DT technique, artificial neural network, etc. Among the algorithms used the study concluded that KNN classifier and DT technique are the best algorithm to use in predicting student performance. However the study also shows that there should be a better understanding on the factors affecting student success therefore the study suggest that the future researches must focus on incorporating more algorithm, using precision and recall metrics, and gathering enough data to improve assessing of the model's performance.[1].

Nackman, G. B., et al., (2022) performed a study to evaluate and improve student skills during the clerkship surgery of 4th year medical students by implementing a self-study CD-ROM educational module. As a result, the implemented self-study CD-ROM improved students test score which means self-study has a significant effect on improving student skills.[2].

The research undertaken by Mahesh Gadhavi and Chirag Patel (2017) discussed the prediction of student's final grade using a univariate linear regression model. The conducted study helped the students to be aware of their final grade on a specific subject and although initially the model focuses on univariate approach they concluded that there is a potential for it to expand into a multivariate framework through incorporating additional parameters to improve the accuracy of the predictions.[3].

El Guabassi, et al., (2021) attempted to build a predictive model to evaluate student's performance by applying several algorithms such as ANCOVA, Logistic regression, support vector algorithm, Log-linear regression, Decision Tree etc..The study aims to evaluate the performance of each algorithm by comparing the accuracy or performance of each. Then the second aim of the study is to determine the relationships between the input features and the target which is the student's academic performance. As a result, they conclude that "Raised Hand","Visited Resources","Parent Responsible","Parent Answering Survey" and "Student Absence Days"are the most significant features for predicting student's academic performance.[4].

Zafari, M., et al., (2021) Developed a Machine Learning (ML) based system that will assess the performance of high school students through identifying the significant factors that influence their academic performance. The system categorizes the students into four classes after evaluating their performance 'very well', 'good', 'medium', and 'bad'. After categorizing the students the model we're trained again with the decreased dataset then the system evaluated the students without human intervention.[5].

Altabrawee, H., et al., (2019) Used four machine learning techniques to build a classifier that predicts a student's performance in a computer science subject. This includes Artificial Neural Network, Decision Tree, Logistic Regression, and Naive Bayes. In this study, the performance of four models are compared to each other through ROC index performance measure. The result indicate that ANN model is the best model among the four as it returns an accuracy of 77.04.[6].

According to the study conducted by Sundary J., et al., (2021), blended learning, study hours, and motivation has a significant effect on the performance of the student of Universitas Negeri Yogyakarta (UNY) and University of Putra Malaysia (UPM). This result is achieved by data collection technique using survey questionnaires and regression test is used as data analysis technique in this study.[7].

Baliyan, S. P., and Khama, D. (2020) Performed a study that analyze the impact of distance to school and hours spent studying after school in student's performance on Mathematics and English subjects.Data collection for this study is performed through survey, the respondents are 168 students taking the mentioned subjects. Therefore, the study revealed that distance to school and study hours affect the student's performance significantly in Mathematics. On the other hand, the said factors did not influence the performance of the students in the English subject.[8].

Suranto, S., and Pramitasari, E. (2024) Explore the impact of effective study hours in the school environment. The research was performed by conducting an in-depth interview with the students, teachers, and vice principal, the results shows that the concept of effective study hours and student's class schedule must be considered when setting teaching and learning activities in order for the students to perform well in their studies.[9].

Mino, Y., et al., (2001) Investigate the effect of a 1 hour educational program to change attitudes towards mental illness. pre- and post- questionnaire design is used to perform this study and the results indicate that the 1 hour educational program greatly influenced student's attitude towards mental illness allowing them to have a better understanding on this topic.[10].

3 Methods

Machine Learning brought a huge development in our modern world particularly education, machine learning algorithms makes it possible to organize task such as scheduling and grading which can give both students and teachers more time to focus on teaching and studying. Moreover, predictive analytics enable to recognize students that needs to improve academically which can prevent dropouts and boost academic performance.

Univariate Linear Regression

A Univariate Linear Regression is one of the important techniques in machine learning as it is used to evaluate the relationship between each independent and dependent variable. In this study, the dependent variable will be used on training the model, the data will be divided into two which are the training and testing set. The training set will be used to train the model and the trained model will be saved using pickle for making a flask application. On the other hand the testing set will be used for predicting the student's score. The evaluation of the trained model will be performed using R-squared in order to assess the model's accuracy on predicting the student's performance based on their study time. Through this, we can analyze the importance of having enough study time and how the time spent by the students studying can affect their academic success

4 Dataset and Features

The dataset was retrieved from Kaggle Repository, it contains only 2 columns which are "Hours" and "Scores", the variable Hours will be the input feature for this model representing the number of hours a student studied, the model will predict the score of the student based on the number of hours they studied. A simple linear regression can be used in this dataset however in this study, a Univariate Linear Regression Approach will be used in order to evaluate if the number of hours a student studied will affect their academic performance.[11]

	count	mean	std	min	25%	50%	75%	max
Hours	25.0	5.012	2.525094	1.1	2.7	4.8	7.4	9.2
Scores	25.0	51.480	25.286887	17.0	30.0	47.0	75.0	95.0

Figure 1: Dataset Description

5 Experiments/Results/Discussion

The study is performed using a Univariate Linear Regression, choosing the dependent variable for training and testing the model. The distribution of scores and relationship between student's study hour and score are visualized for better analysis of the model's process. After training the model, it is saved using pickle to use for flask application. The Actual and Predicted result is then visualized in a comparative way to asses how close the predicted result is from the actual ones. As a result, The linear regression model was able to predict student scores based on study hours with high accuracy. As an example, we set the initial number of study hours of a student to 5 hours/day, the model predicted a score of approximately 51.57% in the exam which implies that the time spent for studying have significant effect on student's score.

	Actual	Predicted
0	20	16.884145
1	27	33.732261
2	69	75.357018
3	30	26.794801
4	62	60.491033

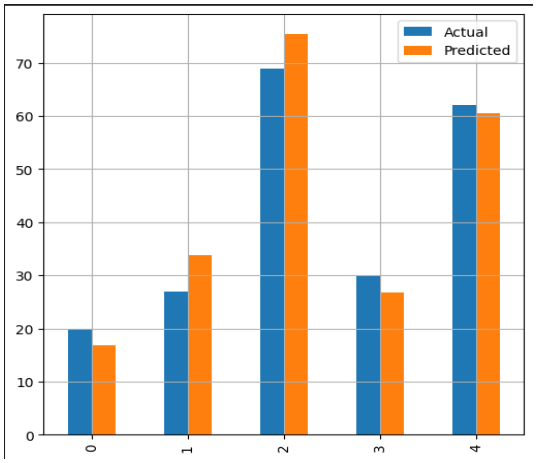

Figure 2: Actual vs. Predicted

Flask Application

A flask framework is developed to ask input from the user to predict the student's grade. HTML is incorporated directly together with the CSS code to improve the visual design of the web application to provide better experience for the user. The figure displays the interactive page of the model wherein the users can input a specific study hours for a student and the model will return the predicted score based on the entered value.

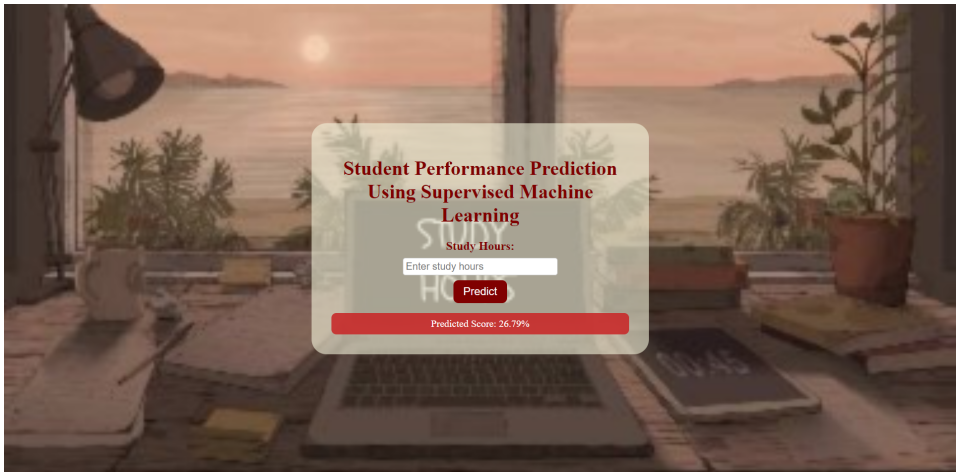

Figure 3: Flask Homepage

6 Conclusion/Future Work

The study successfully demonstrates the use of Univariate Linear Regression on predicting the student's score based on the number of study hours. The result shows an accuracy of 95.16% in training score and 94.55% in testing score, this signify a great performance on predicting students score using the training and testing. Moreover, the evalation of the model's performance in predicting student's grade based on their study hours returned an accuracy of 94.55%. This concludes that the study provided valuable insights into the relationship between study hours and student performance, demonstrating the effectiveness of supervised machine learning in educational research.

The findings in this study can help inform strategies to improve student performance through targeted study practices. Future studies could use larger dataset, a more advanced algorithm, and incorporate more features such as study environment and student motivation to improve the model's prediction of student's performance.

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